Genuine Multipartite Entanglement Analysis

Hubbard ring - Convex-roof framework

System parameters

Model	**Hubbard 1D-PBC**
Mode	**uniform**
Sites (L)	**6**
Electrons (Ne)	**6**
Hopping (t)	**1.0**
Interaction (U)	**4.0**
n-body order	**5**
Eigenstate idx	**0 (ground state)**

Eigenstate properties

Energy E0	**-3.66870618**
Energy E1	**-2.89838147**
Gap (E1-E0)	**0.770325**
<S^2>	**-0.000000**
S	**-0.0000**
<Sz>	**0.000000**
Multiplicity	**1.0**
Spin label	**Singlet (S=0)**

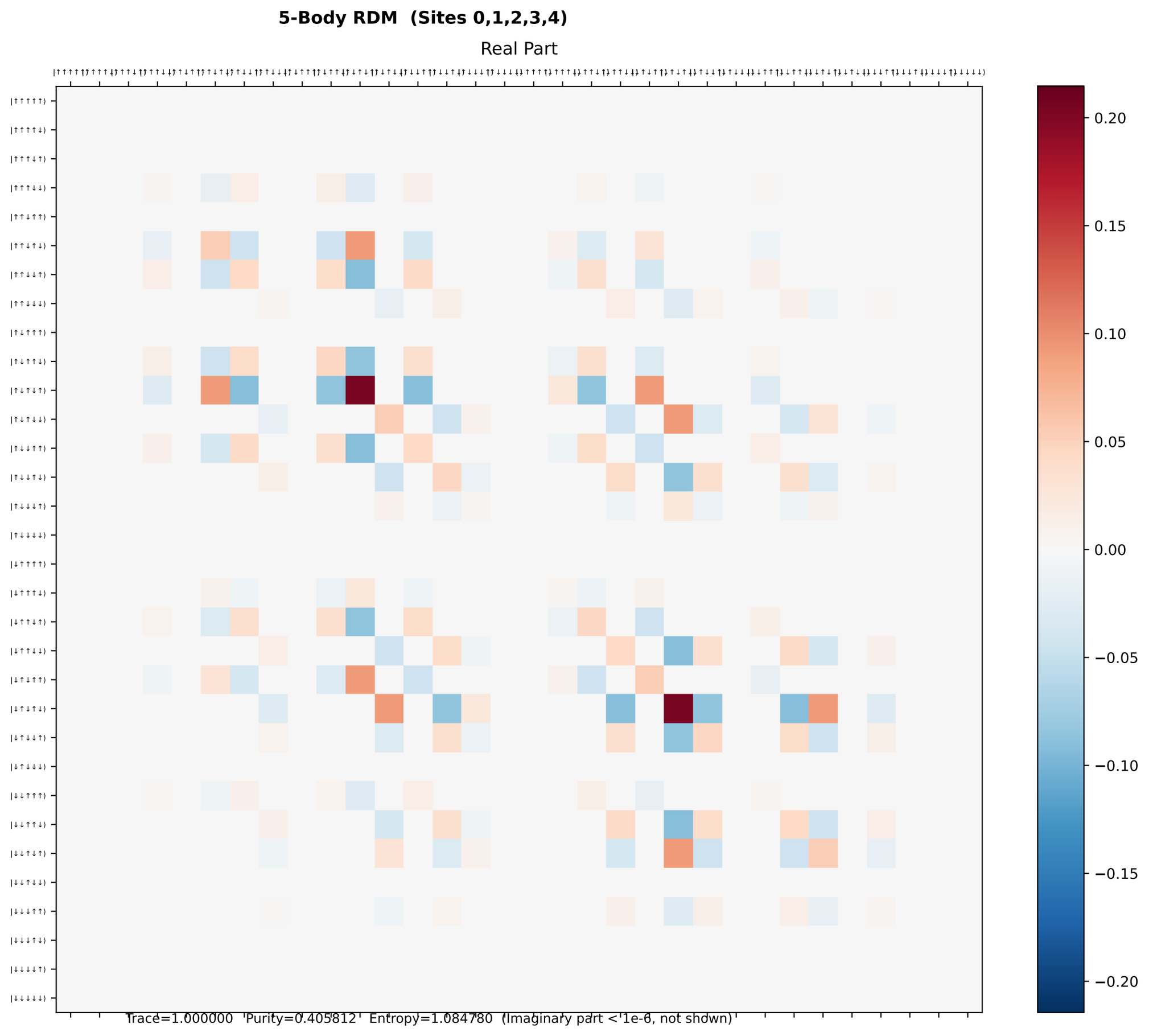
Double occupancy

<D> total	**0.666395**
<D> / site	**0.111066**

RDM and GME statistics for this run (5-body clusters)

Hilbert space dim : 2^5 = 32
Number of clusters : 6
Pure clusters : 0
Mixed clusters : 6
Clusters with GME (>0) : 6

Purity gamma = Tr(rho^2) : mean = 0.405812 [min 0.405812 , max 0.405812]
Von Neumann entropy : mean = 1.084780 [min 1.084780 , max 1.084780]
Rank of rho : mean = 10.00 [min 10 , max 10]
C_GME (best estimate) : mean = 0.402491 [min 0.401901 , max 0.403482]


5-Body RDM (Sites 0,1,2,3,4)
Real Part
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Trace=1.000000 Purity=0.405812 Entropy=1.084780 (Imaginary part <1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 0,1,2,3,4)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4058117098
  Von Neumann entropy  : 1.0847796368
  Eigenvalue min       : -1.065002e-16
  Eigenvalue max       : 4.486714e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.40581171

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

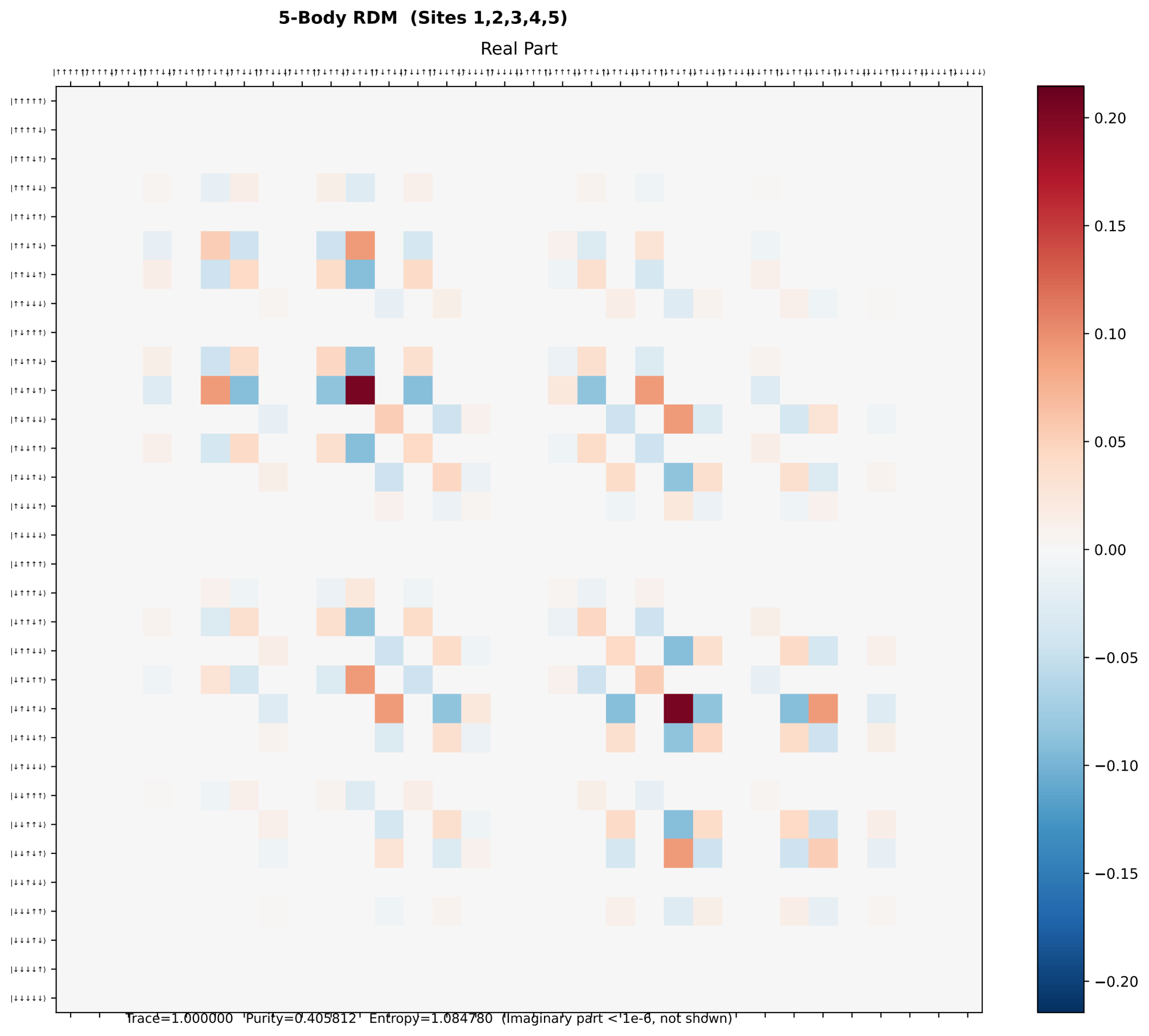
STATUS: MIXED STATE
-----------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
===================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.40246729   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.75739264

  Status: nONE
```


5-Body RDM (Sites 1,2,3,4,5)
Real Part
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Trace=1.000000 Purity=0.405812 Entropy=1.084780 (Imaginary part <1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 1,2,3,4,5)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4058117098
  Von Neumann entropy  : 1.0847796368
  Eigenvalue min       : -6.099756e-17
  Eigenvalue max       : 4.486714e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.40581171

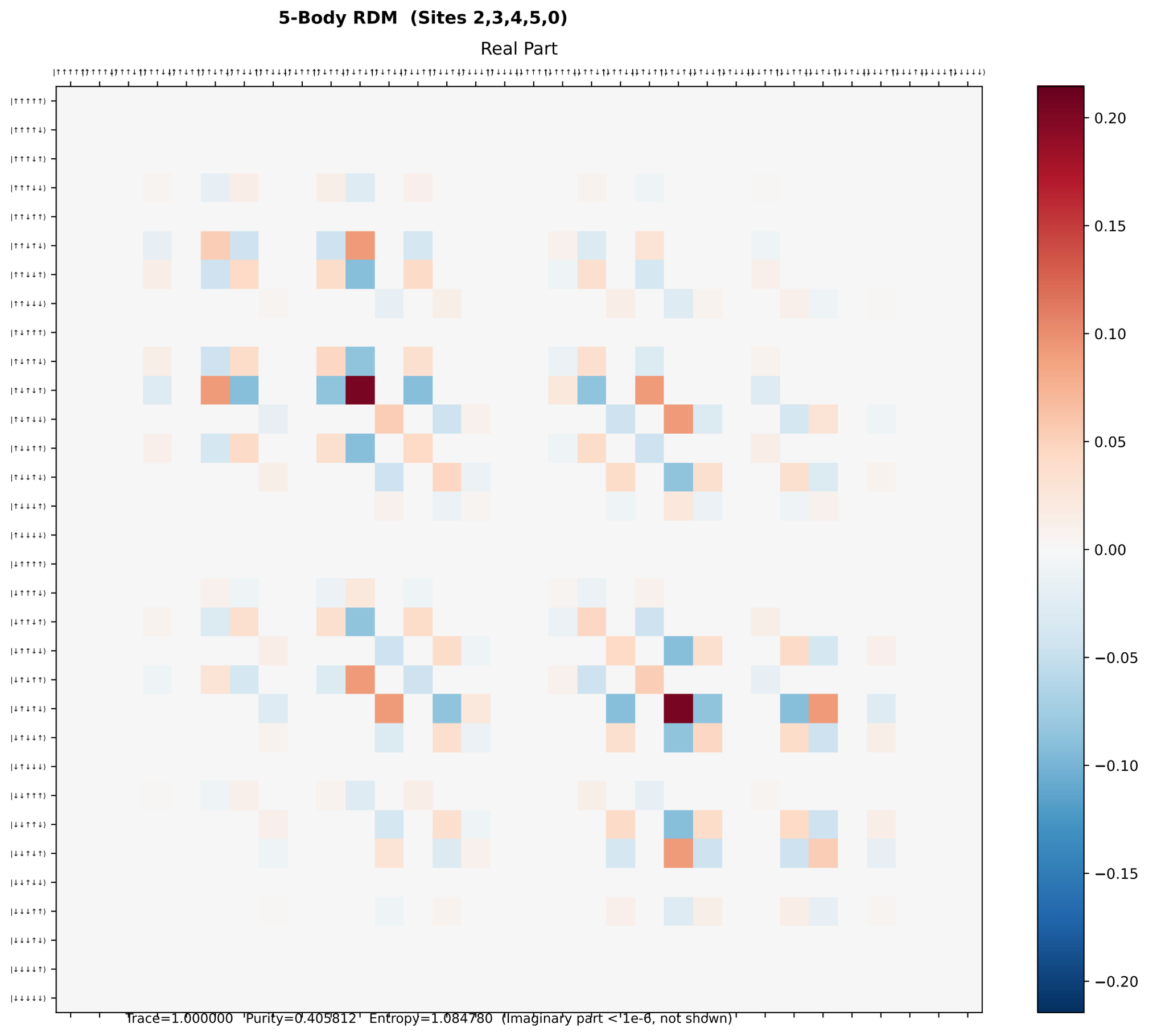
  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------------------------
Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.40190144   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.75739264

  Status: nONE
```


5-Body RDM (Sites 2,3,4,5,0)
Real Part
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Trace=1.000000 Purity=0.405812 Entropy=1.084780 (Imaginary part <1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 2,3,4,5,0)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4058117098
  Von Neumann entropy  : 1.0847796368
  Eigenvalue min       : -2.724496e-17
  Eigenvalue max       : 4.486714e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.40581171

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

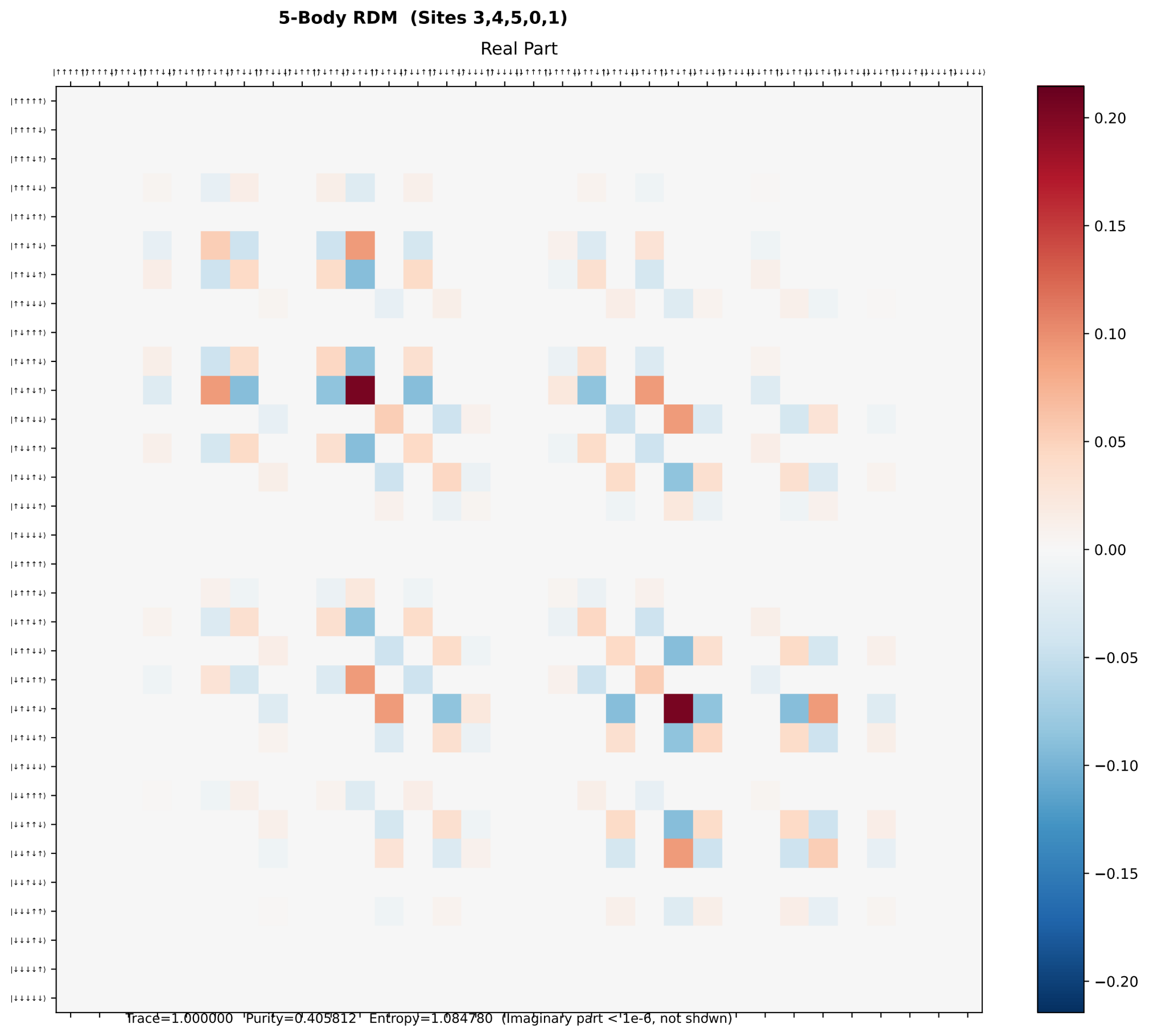
STATUS: MIXED STATE
-----------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
===================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.40247893   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.75739264

  Status: nONE
```


5-Body RDM (Sites 3,4,5,0,1)
Real Part
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Trace=1.000000 Purity=0.405812 Entropy=1.084780 (Imaginary part <1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 3,4,5,0,1)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4058117098
  Von Neumann entropy  : 1.0847796368
  Eigenvalue min       : -3.582718e-17
  Eigenvalue max       : 4.486714e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.40581171

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

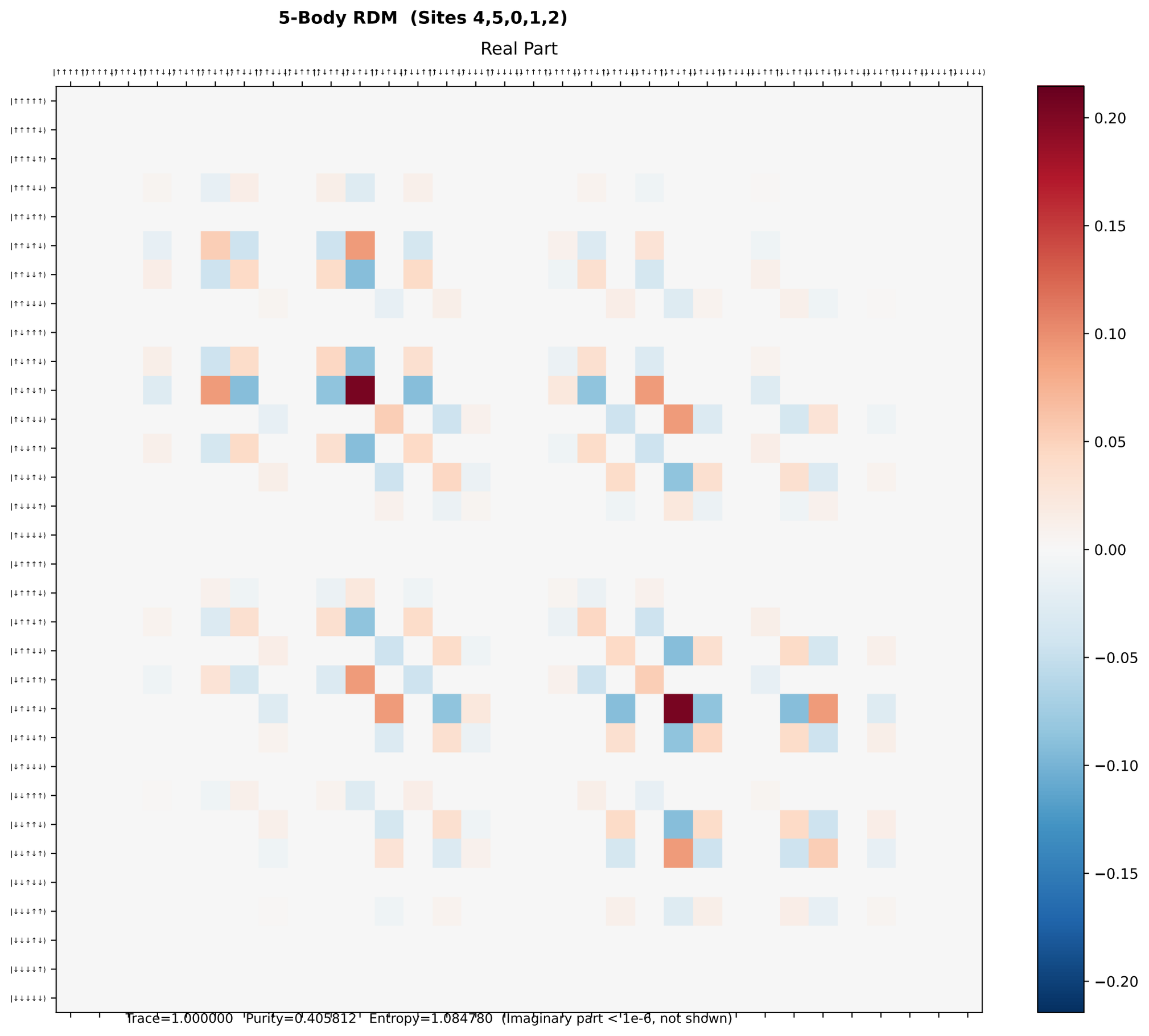
STATUS: MIXED STATE
-------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.40348211   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.75739264

  Status: nONE
```


5-Body RDM (Sites 4,5,0,1,2)
Real Part
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Trace=1.000000 Purity=0.405812 Entropy=1.084780 (Imaginary part <1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 4,5,0,1,2)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4058117098
  Von Neumann entropy  : 1.0847796368
  Eigenvalue min       : -1.999869e-17
  Eigenvalue max       : 4.486714e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.40581171

  A            | B            | Tr(rho_A^2)
  ____________ | ____________ | ___________

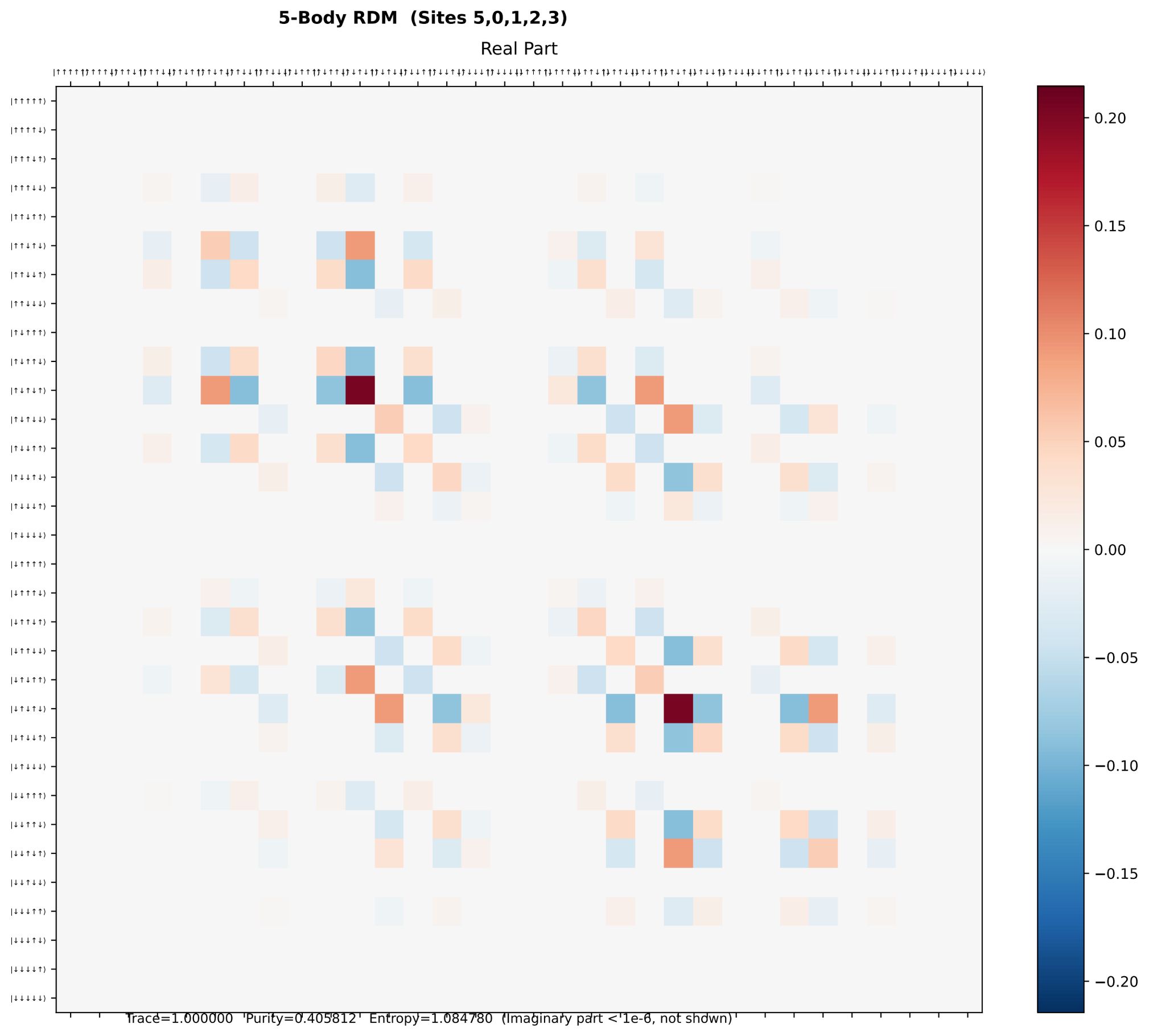
STATUS: MIXED STATE
___________________________________

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.40241179   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.75739264

  Status: nONE
```


5-Body RDM (Sites 5,0,1,2,3)
Real Part
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Trace=1.000000 Purity=0.405812 Entropy=1.084780 (Imaginary part <1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 5,0,1,2,3)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4058117098
  Von Neumann entropy  : 1.0847796368
  Eigenvalue min       : -2.377449e-17
  Eigenvalue max       : 4.486714e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.40581171

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.40220400   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.75739264

  Status: nONE
```